

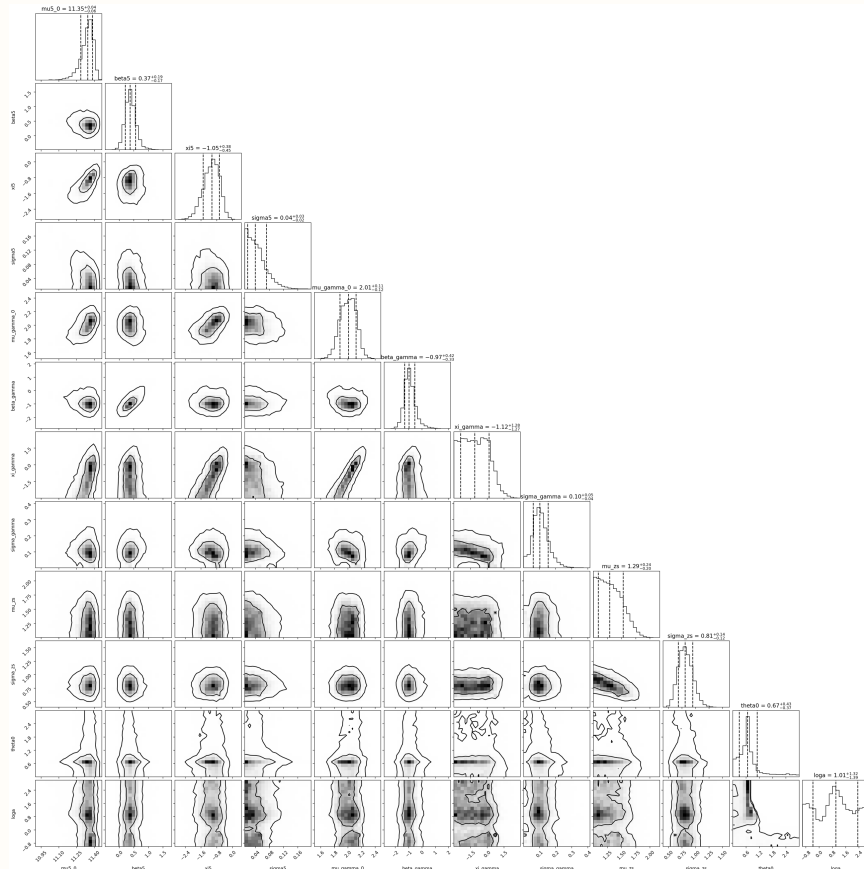
Posterior Corner Plots

Slide 1 / 3

Latest production posterior corners created on 2026-03-11; these runs also feed the canonical PPC and posterior-trend summaries.

de Vaucouleurs

20260308_215643_devauc_devauc_prod_20260308



Profile-specific model note: n is fixed to 4 in the de Vaucouleurs branch.

Sersic

20260308_221211_sersic_sersic_prod_20260308

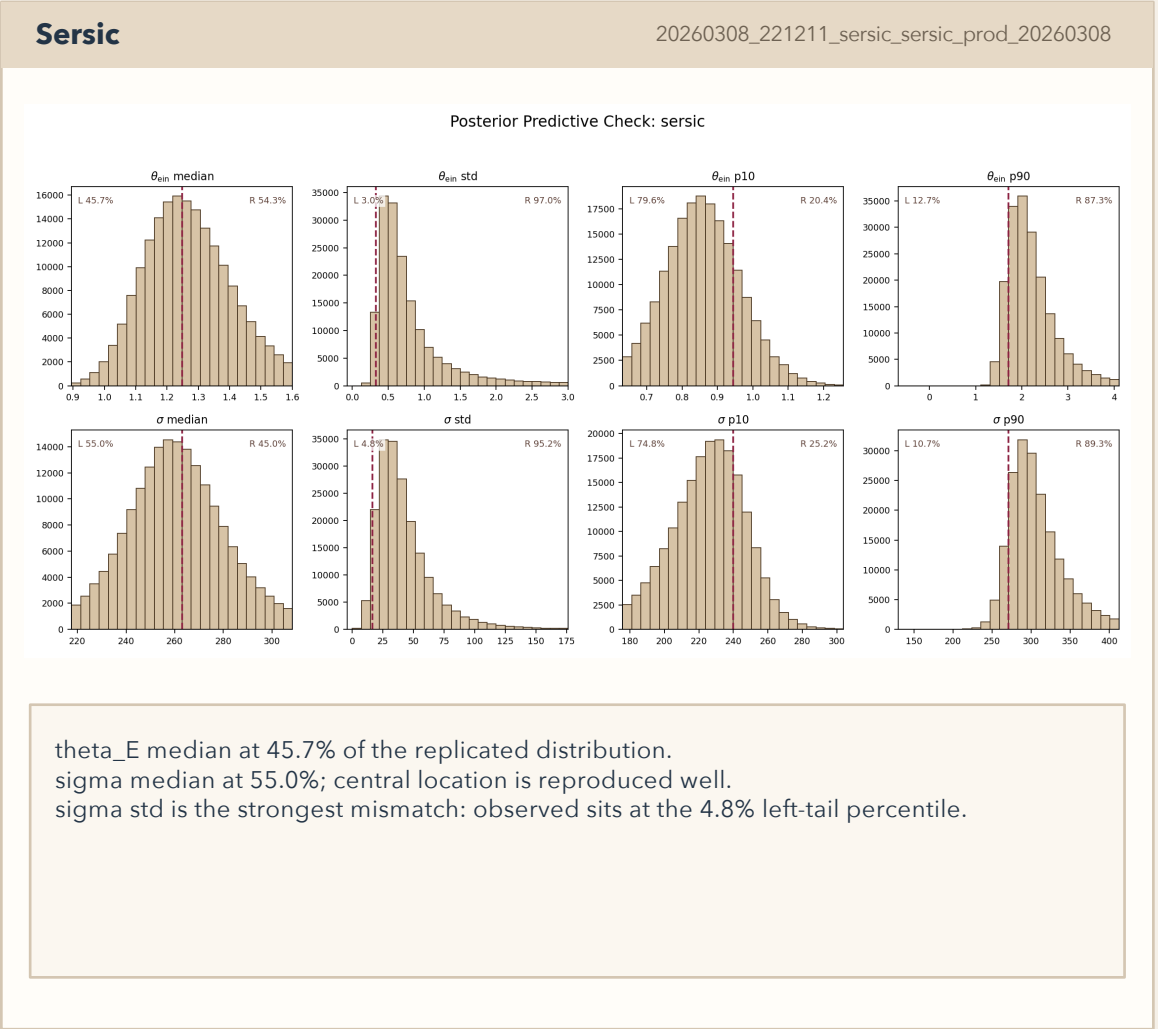
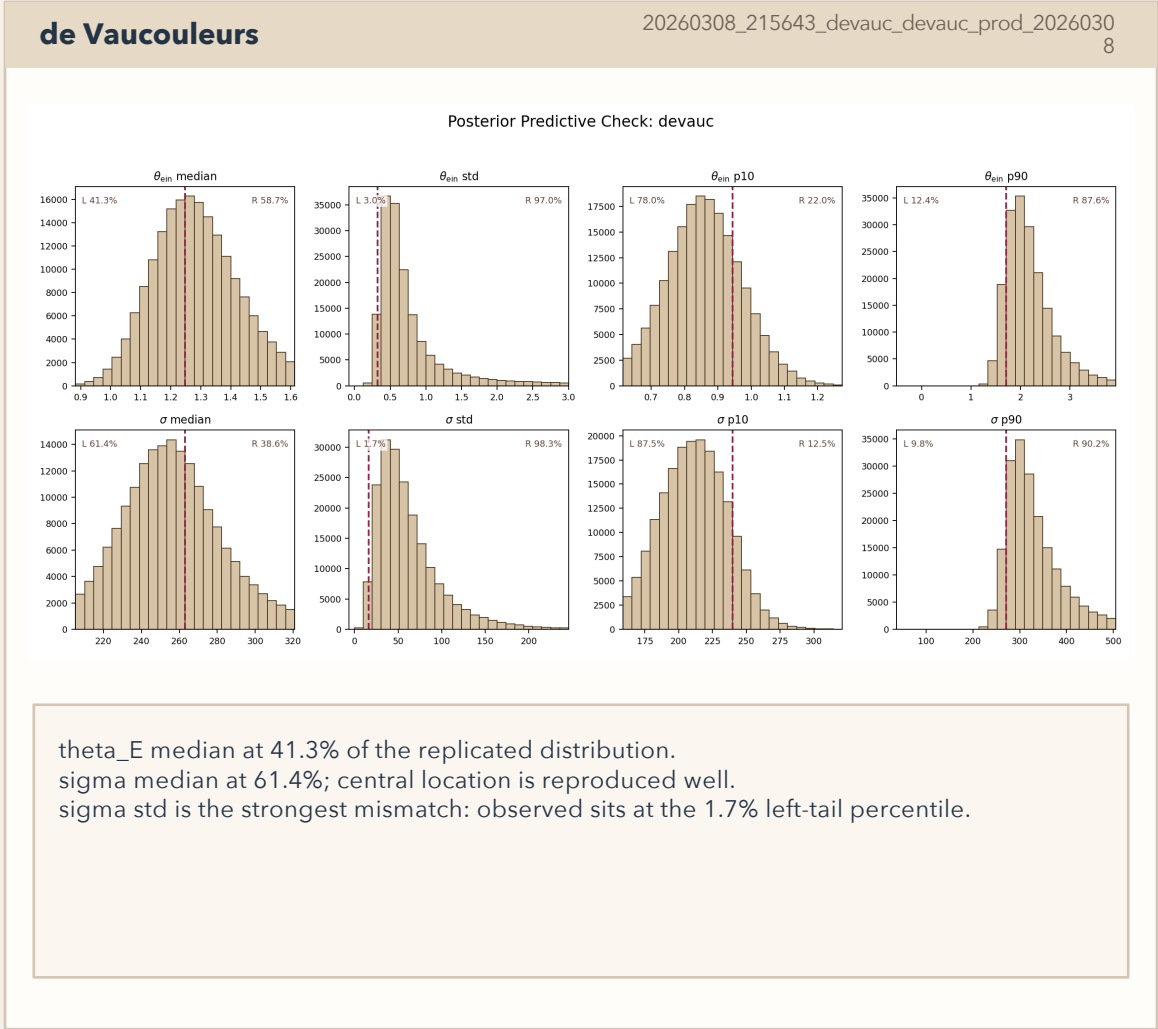


Profile-specific model note: Sersic carries n-dependent structure in the full workflow.

Shared 12D hyper-parameter family; profile difference enters through the light-profile treatment. These exact runs are the sources for the canonical PPC and Fig. 8-like summaries shown next.

Canonical Posterior Predictive Check

2026-03-10 canonical rerun: tail-capped full chain, burn-in 2000, 192000 posterior draws, candidate pool 100000, process-pool workers 12.



Observed medians land near the bulk of the replicated distributions for both profiles. The most persistent tension is the unusually small observed sigma scatter relative to the replicated sigma std panel.

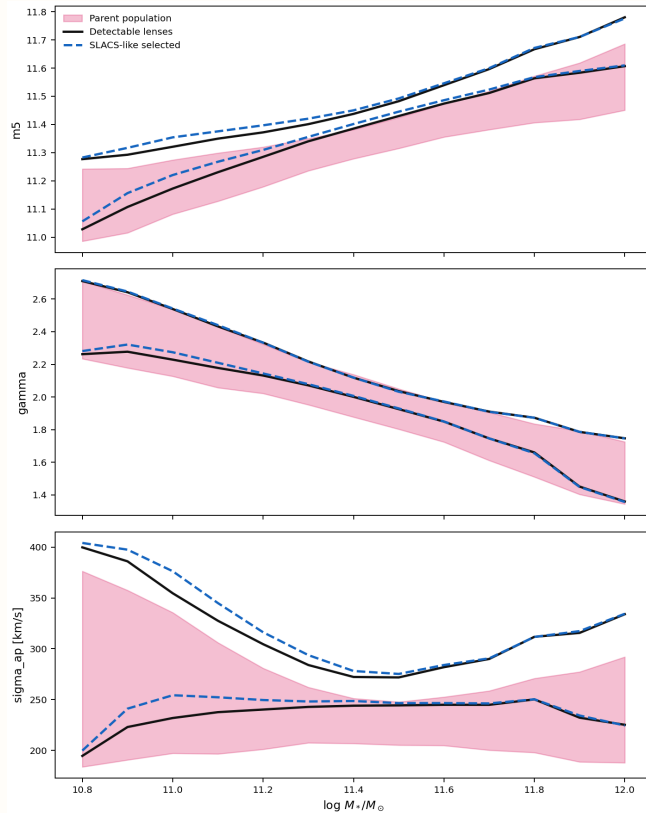
Fig. 8-like Posterior Trends

Slide 3 / 3

Separate evaluator from histogram PPC: fixed mass-grid conditional means for m5, gamma, and sigma_ap with parent / detectable / SLACS-like selected populations.

de Vaucouleurs

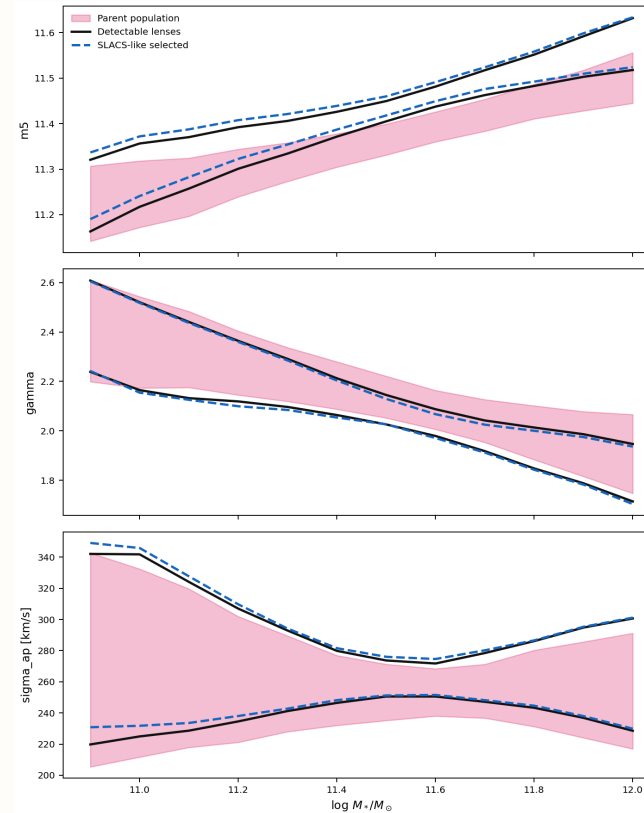
20260308_215643_devauc_devauc_prod_20260308



m5 selection uplift: +0.05 -> +0.12 dex from low to high mass.
gamma selection shift: +0.04 -> +0.02; sign tells whether selection pushes the curve above or below parent.
sigma_ap boost: +24.0 -> +31.1 km/s across the mass range.

Sersic

20260308_221211_sersic_sersic_prod_20260308



m5 selection uplift: +0.03 -> +0.08 dex from low to high mass.
gamma selection shift: +0.01 -> -0.06; sign tells whether selection pushes the curve above or below parent.
sigma_ap boost: +12.0 -> +11.3 km/s across the mass range.

Current production trend contract: $n_{\text{posterior_draws}} = 256$, $n_{\text{mass_grid}} = 40$, $\log M^*$ in $[11.2, 12.1]$, $n_{\text{candidate_per_mass}} = 2000$. Both profiles show rising m_5 and falling γ with mass; selection boosts σ_{ap} in both cases.